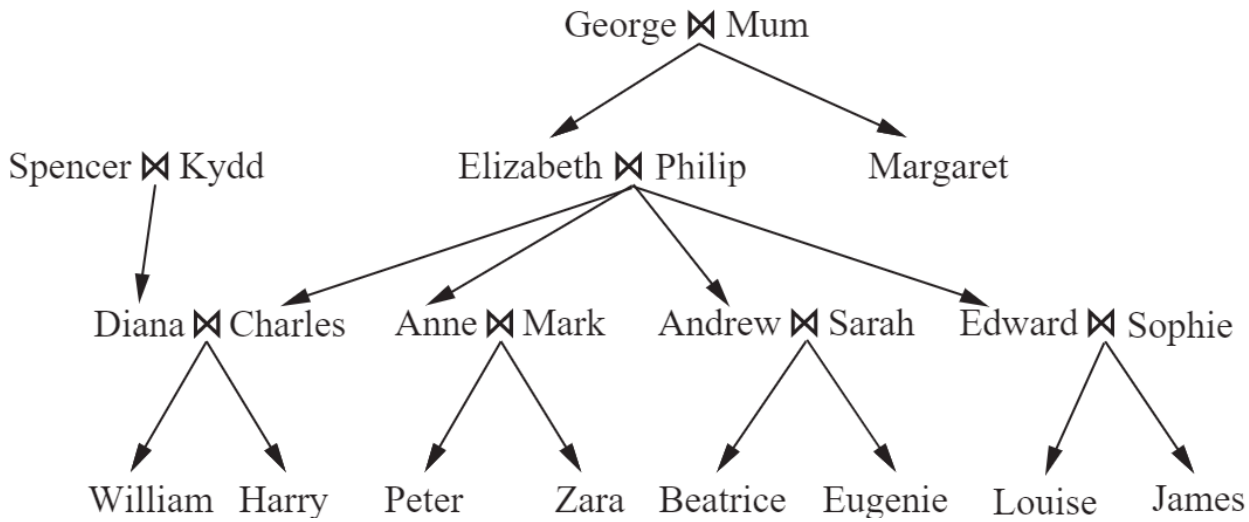
CMSC-4200 Artificial Intelligence
Spring 2024, Group Assignment 6

Due – Monday, May 4th, 2026, at 11:59pm EST

This programming assignment will be graded out of 30 points. Submit your Prolog program to the right D2L Dropbox named "Assignment 6 Dropbox" by the due date and time. At the top of the program, place a comment header containing program name, group number and your name(s).

Consider the family tree below:



The symbol ⋈ connects spouses and arrows point to children.

Write a Prolog program that contains:

a) **Relation Facts** to state the relationships among this family members using only these predicates: *child*, *spouse*, *male*, and *female*.
b) **Relation Rules** to infer other facts about: *sibling(X,Y)*, *mother(X,Y)*, *father(X,Y)*, *brother(X,Y)*, *sister(X,Y)*, *son(X,Y)*, *daughter(X,Y)*, *two_sisters(X,Y)*, *two_brothers(X,Y)*, *three_siblings(X,Y,Z)*, *married(X)*, *grandchild(X,Y)*, *great_grandparent(X,Y)*, *brother_in_law(X,Y)*, *sister_in_law(X,Y)*, *aunt(X,Y)*, *uncle(X,Y)*, and *has_exactly_two_children(X,Y,Z)*.

Your Prolog program should be able to answers questions such as who are George's grandchilders, Diana's brothers-in-law, Zara's great-grandparents, who has exactly two children, and other questions.